



University of the Philippines Los Baños

College of Engineering and Agro-industrial Technology  
Department of Electrical Engineering



# EE 187 Course Pack

## Digital Signal Processing

### *2nd Semester A.Y. 2025-2026*

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# Final Project: Speech Retrieval

## Learning Objectives

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At the end of the final project, the student shall be able to

- Identify digital filter design specifications based on real-world problems
- Design digital filter/s to achieve requirements

## Design Specifications

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Each group (maximum of two members) will be provided with four speech signals: two clean vocals and two distorted vocals. The groups are tasked with implementing digital filtering techniques on the distorted vocals to retrieve the speech information. The retrieved speech signals will then be compared with the corresponding clean vocals to determine whether the required specifications have been met. Groups may use any DSP techniques learned in the course to recover the speech information.

## Requirements and Grading

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The final project will be graded based on the following criteria:

### Digital Filter Design and its Functionality (60%)

The digital filter design must comply with the required features stated in the design specifications. Failure to meet these specifications will result in point deductions. The following grading rubric will be used:

| Grade | SNR (dB) | MSE (%) | Grade | SNR (dB) | MSE (%)  |
|-------|----------|---------|-------|----------|----------|
| 100%  | [0-1)    | [0-10)  | 50%   | [5-6)    | [50-60)  |
| 90%   | [1-2)    | [10-20) | 40%   | [6-7)    | [60-70)  |
| 80%   | [2-3)    | [20-30) | 30%   | [7-8)    | [70-80)  |
| 70%   | [3-4)    | [30-40) | 20%   | [8-9)    | [80-90)  |
| 60%   | [4-5)    | [40-50) | 10%   | [9-max)  | [90-100) |

*Note: (1) In case SNR is less than 9, the grading will be scaled based on these rubrics. (2) MSE can be computed by dividing  $MSE_{filtered}/MSE_{distorted}$*

### Final Report (30%)

Proper documentation of the digital filter design must be submitted in the form of a written report. The report must include the following:

- Methodology
  - This section should answer the following questions:
    - How was the digital filter designed?
    - What factors were considered in selecting the DSP technique(s)?
  - Include supporting figures, diagrams, and analysis where appropriate.
- Results and Discussion
  - This section should answer the following questions:
    - Did the designed digital filter meet the required specifications? Why or why not?

- What problems and limitations were encountered during the digital filter design process?
- References
  - List all references used in the preparation of the report.
  - Please note that AI tools may be used as guides but should not be cited as primary references.

### **Peer evaluation (10%)**

A multiplier (0.0-1.0) based on peer evaluation will be applied to the total points. A link to the peer evaluation will be provided after the submission of the final project.

## **Final Notes**

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Students should consult frequently during the laboratory hours and FIC's consultation hours.

This guide may be updated as needed. See announcements in Google Classroom for recent changes.



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